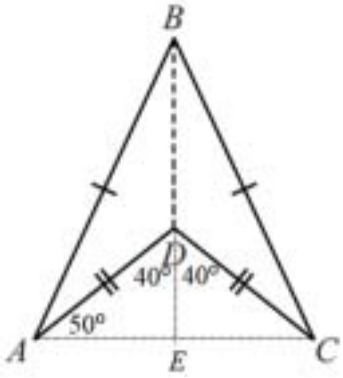
Question	Answer	Marks	Partial Marks
4(b)	$\frac{-1\pm\sqrt{1^2-4(3)(-6)}}{2(3)}$ Or $-\frac{1}{6}\pm\sqrt{2+\left(\frac{1}{6}\right)^2}$ oe or $\frac{1}{3}\left(-\frac{1}{2}\pm\sqrt{18+\left(\frac{1}{2}\right)^2}\right)$ oe	**B2**	**B1** for $\sqrt{1^2-4(3)(-6)}$ or for $\frac{-1+\sqrt{p}}{2(3)}$ or $\frac{-1-\sqrt{p}}{2(3)}$ or $\left(r+\frac{1}{6}\right)^2$ or $\left(3r+\frac{1}{2}\right)^2$
	–1.59 and 1.26	**B1**	
4(c)	9.028 to 9.040	**2**	**M1** for (*their* root (greater than –1) + 1) × 4
5(a)(i)	3 cao	**1**	
5(a)(ii)	–2, –0.45 to –0.4, 2.40 to 2.45	**3**	**B1** each
5(a)(iii)	3 cao	**1**	
5(a)(iv)	Asymptote	**1**	
5(b)(i)	Correct ruled line	**2**	**B1** for ruled line through (0, –2) but not $y = -2$ or for ruled line with gradient 1
5(b)(ii)	1 cao	**1**	
5(c)	Substituting values of x and y into $y = x^2 - \frac{c}{x}$ for an exact point on graph of $y = f(x)$ or substituting *their* value of x from 5b(ii) into $x^2 - \frac{c}{x} = x - 2$	**M1**	
	leading to $c = 2$ with no errors	**A1**	
5(d)	[p =] –1 and [q =] 2 nfww	**2**	**M1** for $x^3 - x^2 + 2x = 2$ seen or **B1** for each nfww
6(a)	4.27 or 4.272...	**2**	**M1** for $4^2 + 1.5^2$ oe

Question	Answer	Marks	Partial Marks
6(b)	203 or 202.6…	**3**	**B2** for [angle at W =] 22.6... or for [angle at V =] 67.4 or 67.38… or **M1** for tan = $\frac{5}{12}$ or $\frac{12}{5}$ oe
6(c)	25.2 or 25.20 to 25.21[0]  	**5**	**B4** for [BC or AB =] 7.6[0] or 7.604 to 7.605 OR **M3** for a complete explicit method leading to AB or BC, e.g. $\frac{5\sin 140}{\sin 25}$ OR **M2** for a complete implicit method leading to AB or BC, e.g. $\frac{\sin 25}{5} = \frac{\sin 140}{BC \text{ or } AB}$ oe and **M1** (dep on AB from trig) for 2 × *their* AB + 10 OR **B1** for any relevant angle E.g. $\angle BDA$ or $\angle BDC$ = 140, $\angle DAE$ or $\angle DCE$ = 50 or $\angle ADE$ or $\angle CDE$ = 40 or $\angle ADC$ = 80
6(d)	79.5 or 79.6 or 79.54 to 79.55...	**5**	**B2** for [PR^2 =] 245 or 245.1 to 245.2 or [PR =] 15.65 to 15.66 or 15.7 or **M1** for [PR^2 =] $11^2 + 8^2 - 2 \times 11 \times 8 \times \cos 110$ **M2** for [cosPQR =] $\frac{10^2 + 14^2 - (\textit{their } PR)^2}{2 \times 10 \times 14}$ oe or **M1** for (*their* PR)^2 = $10^2 + 14^2 - 2 \times 10 \times 14\cos PQR$ oe
7(a)(i)	40	**2**	**M1** for $\frac{50}{75}$ [× 60] oe
7(a)(ii)	36 nfww	**3**	**M2** for $\frac{47 - 0.5}{75 \text{ to } 80}$ [× 60] or $\frac{46 \text{ to } 47}{75 + 2.5}$ [× 60] or **M1** for 47+0.5 or 47 – 0.5 or 75 + 2.5 or 75– 2.5